FIITJEE REVIEW MOCK TEST-03

(BATCHES: 921 LOT)

IIT – JEE 2021

Time: 1 Hours **Maximum Marks: 69**

A. Question Paper Format

1. The question paper consists three parts.
2. **Part A – (I)** contains **5** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part A – (II)** contains **5** multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part B** contains **2** questions. Each question has four statements (A, B, C and D) given in column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with **one or more** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for that particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.
5. **Part C** contains **6** questions. The answer to each of the questions is a **single – digit integer,** ranging from 0 to 9.

B. Marking scheme:

5. For each question in Part A (I), you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.
6. For each question in Part A (II), you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.
   Partial Marks: **+1** For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
   Zero Marks: **0** If none of the bubbles is darkened.
   ***Negative Marks*: −2 In all other cases.**
   For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in −**2** marks, as a wrong option is also darkened.
7. For each question in **Part B**, you will be **awarded 2 marks** for **each row** in which you have darkened the bubble(s) corresponding to the correct answer. Thus, each question in this section carries a maximum of **8** marks. There is **no negative marking** for incorrect answer(s) for this section.
8. For each question in **Part C**, you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **Date:**. . . . . .

PHYSICS

PART – A(I) (Single Correct Choice Type)

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. A silicon diode requires minimum current of 2mA to be above its knee point of its (V-I) characteristics. Assuming voltage across diode is constant above its knee point. The value of maximum resistance that should be connected in series with diode and battery of 10 V such that diode operates above its knee point. (Assume resistance of diode to be zero)
   (A) 5 kΩ (B) 5.35 kΩ
   (C) 4.65 kΩ (D) 9.30 kΩ

2. The desirable properties of making permanent magnets are
   (A) high retentivity and high coercive force
   (B) high retentivity and low coercive force
   (C) low retentivity and high coercive force
   (D) low retentivity and low coercive force

3. A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It
   (A) will stay in north-south direction only
   (B) will stay in east-west direction only
   (C) will become rigid showing no movement
   (D) will stay in any position.

4. A heavy string of mass m hangs between two fixed points A and B at the same level. The tangents to the string at A and B are at an angle θ with the horizontal as shown in the figure. The tension in the string at lowest point is



   (A) $\frac{mg}{2\sin\theta}$ (B) $\frac{mg}{2\cos\theta}$
   (C) $\frac{mg}{2\tan\theta}$ (D) $\frac{mg}{2\cot\theta}$

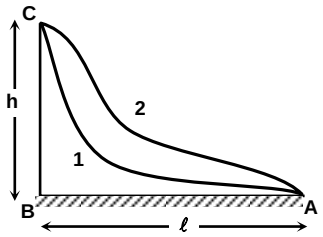
5. The muscles of a normal eye are least strained when the eye is focused on an object
   (A) far away from the eye
   (B) very close to the eye
   (C) at about 25 cm from the eye
   (D) at about 1 m from the eye

Space for rough work

PART – A(II) (Multiple Correct Choice Type)

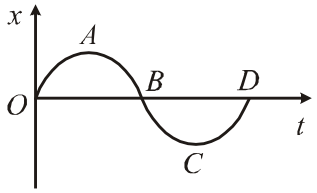
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

6. A body is lifted quasi statically to the top of a mountain through path 1 and path 2 by applying a tangential force from bottom in the figure. Coefficient of friction between the body and surface is μ then



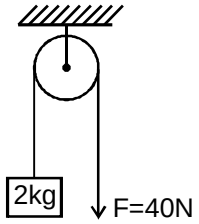
(A) Work done by gravity and friction is less through path 1 than 2.
(B) Work done by gravity and friction is more through path 1 than 2.
(C) Work done by gravity is same in both path.
(D) Work done by friction is same in both path.

7. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements are true with respect to the motion



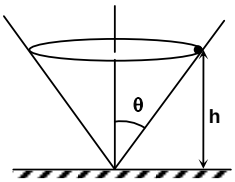
(A) in the motion between O and A the velocity is positive and acceleration is negative
(B) between A and B the velocity and acceleration are positive
(C) between B and C the velocity is negative and acceleration is positive
(D) between C and D the acceleration is positive

8. A block of mass 2 kg is hanging over a smooth and light pulley through a light string. The other end of the string is pulled by a constant force F = 40 N. At t = 0 the system is at rest as shown. Then in the time interval from t = 0 to t = $\frac{2}{\sqrt{10}}$ seconds, pick up the correct statement (s) : (g = 10 m/s^2)



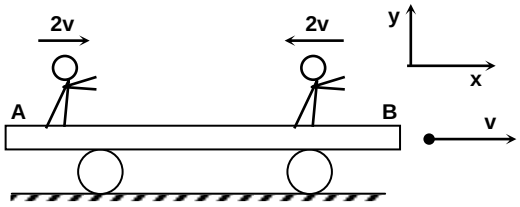
(A) tension in the string is 40 N
(B) work done by gravity is – 20 J
(C) work done by tension on block is 80 J
(D) None of these

9. A particle of mass 'm' is moving in horizontal circle inside a smooth inverted fixed vertical cone above height 'h' from apex. Angle of cone is θ then :



(A) Normal force on particle by surface of cone is mg cos θ
(B) Normal force on particle by surface of cone is mg cosecθ
(C) Time period of revolution of particle increase if θ increase keeping h constant
(D) Time period of revolution increase if h increase keeping θ fixed

10. In a situation, a board moving with a velocity $v\hat{i}$ w.r.t. earth, while a man A is running with a velocity $2v\hat{i}$ w.r.t. earth and the man B is running with a velocity $-2v\hat{i}$ with respect to earth. Both men are running from the opposite ends of the board at the same time as shown. Length of the board is L and they meet after time T, then

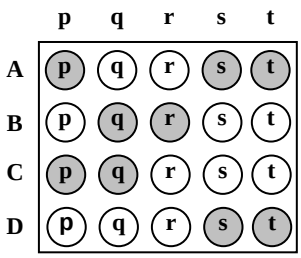


(A) value of T is L/4v
(B) value of T is L./2v
(C) displacement of man B w.r.t. board in time T is 3L/4
(D) displacement of man B w.r.t. board in time T is L/4

Space for rough work

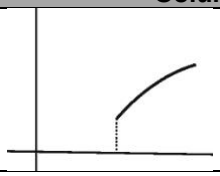
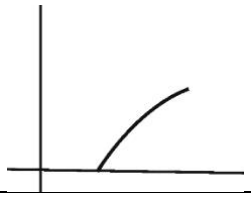
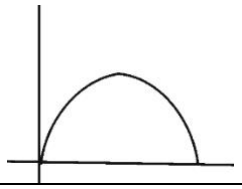
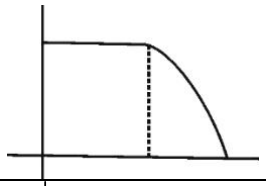
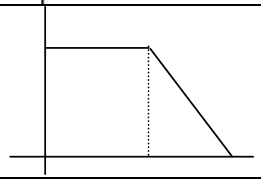
PART – B
Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:



If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:

1. A block of mass m is kept on rough inclined plane of inclination θ. The angle of inclination varies from $\theta = 0$ to $\theta = \frac{\pi}{2}$ very slowly then match the **column I** with **column II.**

	Column–I		Column-II
(A)	Graph of net reaction force by incline(on y-axis) on the block v/s θ(on x-axis) ($\mu_s = \mu_k$)	(p)	
(B)	Graph between acceleration of block (on y-axis) v/s θ(on x-axis) will ($\mu_s = \mu_k$)	(q)	
(C)	Graph between magnitude of friction (on y-axis) v/s θ(on x-axis) ($\mu_s = \mu_k$)	(r)	
(D)	Graph between magnitude of acceleration (on y-axis) v/s θ(on x-axis) ($\mu_s > \mu_k$)	(s)	
		(t)	

Space for rough work

2. There are some situation given in column 1. Match the possible answers in column 2.

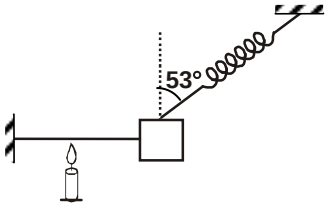
COLUMN – I		COLUMN – II	
(A)	Uniform circular motion of a particle	(p)	Variable acceleration
(B)	Elliptical path of a particle with constant speed	(q)	Variable velocity
(C)	Particle is moving along a circle with variable speed $v = kt, t$ is time and k is positive constant	(r)	Variable centripetal acceleration
(D)	Particle is moving along a curve $x = kt^2$; $y = kt$ in xy plane, where t is time	(s)	Acceleration of constant magnitude
		(t)	Constant acceleration

Space for rough work

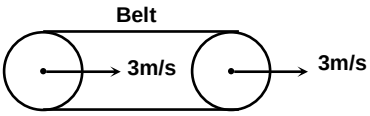
Part C
Integer Answer Type

This section contains **6 questions**. The answer to each of the questions is a **single digit integer**, ranging from **0 to 9.**

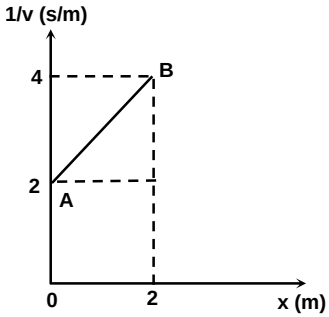
1. The block shown in the figure is in equilibrium. The acceleration of the block just after the string burns is $\frac{ng}{3}$. Find the value of n.



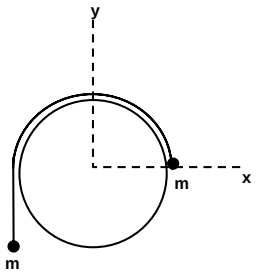
2. A belt of mass 2 kg is wrapped over two cylindrical wheels of small radii rolling on a horizontal rough surface. Each wheel has velocity 3 m/s (see in the figure) and assume that the belt does not slip. If kinetic energy of the belt is 2n joule. Find the value of n.



3. The 1/v versus displacement graph of a particle is shown in the figure, where v is the speed of the particle. The particle is moving in a straight line along positive x-axis. Calculate the time taken (in seconds) by the particle to reach from the point A to B .



4. Figure shows a smooth cylindrical pulley of radius R with centre at origin of co-ordinates. An ideal thread is thrown over it on the two parts of ideal thread two identical masses are tied initially at rest with co-ordinates (R, 0) and (-R, -R) respectively. If mass at x-axis is given a slight upward jerk, it leaves contact with pulley at (R cosϕ, Rsinϕ). Then find ϕ/sinϕ.



5. A simple microscope is rated 5X for a normal relaxed eye. What will be its magnifying power for a relaxed farsighted eye whose near point is 40 cm?

6. A man can adjust power of his eye lens between 50 D and 75 D. His far point is infinity. The distance between his retina and eye lens (in cm) is

Space for rough work

FIITJEE REVIEW MOCK TEST-03

(BATCHES: 921 LOT)

IIT – JEE 2021

ANSWERS

PHYSICS

1.	**C**	2.	**A**	3.	**D**	4.	**C**
5.	**A**	6.	**C, D**	7.	**A, C, D**	8.	**A, C**
9.	**B, C, D**	10.	**A, C**				

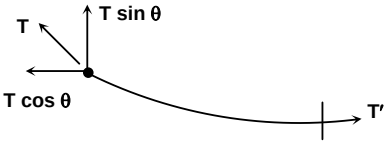
1. **(A) → (s), (B) → (q), (C) → (r), (D) → (p)**
2. **A – (p,q,r,s) (B) – (p,q,r) (C) – (p, q, r) (D) – (q, r, s, t)**

1.	**4**	2.	**9**	3.	**6**	4.	**2**
5.	**8**	6.	**2**				

Solutions

1. $\frac{10-0.7}{2\times10^{-3}} = 4.65\,k\Omega$

4. $T \sin\theta = mg/2$
$T' = T\cos\theta$
$= \frac{mg}{2}\cot\theta = \frac{mg}{2\tan\theta}$



7. Slope of displacement time graph is velocity and rate of change of velocity is acceleration.

8. Acceleration of block is = 10 m/s^2
$\Rightarrow$ Displacement s = $\frac{1}{2}at^2 = \frac{1}{2} \times 10 \times \frac{4}{10} = 2m$
Tension in the string is 40 N

Part C

1. $a = \frac{4}{5}\frac{kx}{m}; \quad \frac{3}{5}kx = mg$

2. K.E. = $\frac{1}{2}\left(\frac{m}{2}\right)(2v)^2 = mv^2 = 18$

3. Area under graph = $\int \frac{1}{v}dx = \Delta t$

4. $mgR\phi - mg\,R\sin\phi = ½\,(2m)v^2$ …. (i)
$mv^2/R = mg\sin\phi$

6. When eye is fully relaxed focal length is largest
$f_{min} = \frac{1}{50}m = 2$ cm
Distance between eye lens and retina is 2 cm.

FIITJEE REVIEW MOCK TEST-03

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

Time: 1 Hour **Maximum Marks: 69**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-05)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (06-10)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are** correct.
4. **Part-B – (11-12)** contains **2 multiple choice questions.** Each question has matching column(s). The codes for the Column(s) have choice (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct
5. **Part-C – (13-18)** contains **06 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 1 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks: – 2In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part B (4),** you will be **awarded +8 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks : + 2 For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.

Other Cases : 0 In all other cases.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **.Session :**. . . . . **Date:**. . . . . .

PART – A(I): (Single Correct Answer Type)

*This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. Conjugate base of $H_2PO_4^-$ is
   (A) H_3PO_4 (B) $H_2PO_4^-$
   (C) HPO_4^{2-} (D) PO_4^{3-}

2. If the equilibrium constant of the reaction of weak acid HA with strong base is 10^9, then the pH of 0.1M NaA is
   (A) 5 (B) 9
   (C) 7 (D) 8

3. Pure ammonia is placed in a vessel at a temperature where its dissociation is appreciable. At equilibrium
   (A) k_p does not change significantly with pressure
   (B) α does not change with pressure
   (C) concentration of NH_3 does not change with pressure
   (D) concentration of H_2 is less than N_2

4. At a certain temperature and a total pressure of 10^5 Pa, iodine vapour contains 40% by volume of iodine atoms $I_2 \rightleftharpoons 2I(g)$. K_p for the equilibrium is
   (A) 10^5 Pa (B) 2.67×10^4 Pa
   (C) 3.22×10^4 Pa (D) 1.52×10^5 Pa

5. Solubility of calcium phosphate (molecular mass M) in water is w per 100 ml at 25°C. Its solubility product at 25°C will approximately
   (A) 10^9 $(w/m)^5$ (B) 10^7 $(w/m)^5$
   (C) 10^5 $(w/m)^5$ (D) 10^3 $(w/m)^5$

Space for rough work

PART – A(II): (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

6. If the concentration of two acids are same their relative strengths can be compared by

(A) $\frac{\alpha_1}{\alpha_2}$ (B) $\frac{k_{a_1}}{k_{a_2}}$

(C) $\sqrt{\frac{k_{a_1}}{k_{a_2}}}$ (D) $\frac{[H^+]_1}{[H^+]_2}$

7. The reaction $SO_2 + Cl_2 \rightleftharpoons SO_2Cl_2$ is exothermic and reversible. A mixture of SO_2(g) and SO_2Cl_2(*l*) is at equilibrium in a closed container. Now a certain quantity of extra SO_2 is introduced into the container, the volume remaining the same. Which of the following are not true
(A)The pressure inside the container will not change
(B)The temperature will not change
(C) The temperature will increase
(D) The temperature will decrease

8. Which of the following factors will favour the backward reaction

$Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g) \quad \Delta H = -ve$

(A) Addition of inert gas at constant pressure (B) Addition of Cl_2 gas
(C) Increase in temperature of reaction (D) Increasing the volume of the container

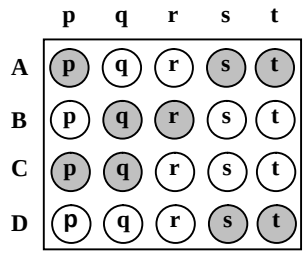
9. In which of the following solvents are not silver chloride most soluble
(A) 0.1 mol dm^{-3} $AgNO_3$ solution (B) 0.1 mol dm^{-3} HCl solution
(C) H_2O (D) Aqueous ammonia

10. Which of the following are a general characteristic of equilibria involving physical processes?
(A) Equilibrium is possible only in a closed system at a given temperature
(B) All measurable properties of the system remain constant
(C) All the physical processes stop at equilibrium
(D) The opposing processes occur at the same rate and there is dynamic but stable condition

Space for rough work

PART – B: (Matrix–Match Type)

*This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II.** The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:*
If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following:



11. Match the following columns :

Column-I		Column-II (At equilibrium)	
(A)	$N_2O_4(g) \rightleftharpoons 2NO_2(g)$ On increasing volume at constant temperature	(p)	Density of gaseous mixture remain constant
(B)	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ On addition of inert gas He at constant P and T	(q)	Density of gaseous mixture decreases
(C)	$2NO(g) + Br_2(g) \rightleftharpoons 2NOBr(g)$ On decreasing pressure at constant temperature	(r)	Average molar mass of gaseous mixture increases
(D)	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ On increasing temperature at constant volume	(s)	Total number of moles of gases increases
		(t)	Value of equilibrium constant increases

12. Note: log 2 = 0.3: All data are at 25^0C

Column-I		Column-II	
(A)	1 M XOH(aq) [pK_b (XOH) = 6]	(p)	$[H^+] > [OH^-]$
(B)	0.1 M $C_6H_5NH_2$(aq) [$pK_b(C_6H_5NH_2)$=7]	(q)	pH = 2.4
(C)	At equivalence point when 20 ml of 0.5 M BOH is titrated with 2M HCl $\left[K_b(BOH) = \frac{1}{4}\times 10^{-9}M\right]$	(r)	pH = 3
(D)	20 ml 0.125 M, C_5H_5NHCl + 5ml, 5×10^{-3} M HCl [K_b (C_5H_5N) = 5×10^{-10} M]	(s)	pH = 10
		(t)	pH + pOH = 14

Space for rough work

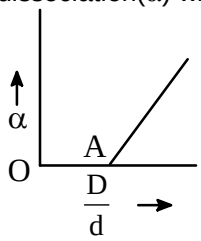
PART-C (Numerical Based Type)

*This section contains **6** questions. The answer to each question is a single-digit integer, ranging from 0 to 9.*

13. For the equilibrium $AB(g) \rightleftharpoons A(g) + B(g)$ at given temperature 1/3rd of AB is dissociated then p/k_p will be numerically equal to …….

14. The reaction takes place in two steps with equilibrium constant 10^{-2} for slow step and 10^{2} for fast step. The equilibrium constant of the overall reaction will be

15 Before equilibrium is setup for the chemical reaction $N_2O_4 \rightleftharpoons 2NO_2$, vapour density(d) of the gaseous mixture was measured. If D is theoretical value of vapour density, variation of degree of dissociation(α) with D/d is as shown in the graph. The value of D/d at the point A is



16. $NH_4COONH_2(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$, if equilibrium pressure is 3 atm for the above reaction, K_p for the reaction is

17. $M(OH)_x$ has k_{sp} of 4×10^{-9} and its solubility 10^{-3} M, the value of x is …….

18. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10}. The pH of the buffer is

Space for rough work

FIITJEE REVIEW TEST-03

CHEMISTRY

(Batches: 921 Lot)

IIT-JEE, 2021

ANSWER KEYS

1. **C**	2. **B**	3. **A**	4. **B**
5. **B**	6. **A, C, D**	7. **A, B, D**	8. **A, C, D**
9. **A, B, C**	10. **A, B, D**		

11. **(A→q, s); (B→q, s); (C→q, s); (D→s, t)**

12. **(A→r, t); (B→s, t); (C→p, q, t); (D→p, t)**

13. **8**	14. **1**	15. **1**	16. **4**
17. **2**	18. **4**		

HINTS & SOLUTIONS

1. C

2. B

3. A
k_p is constant and does not change with pressure.

4. Partial pressure of I atoms $(P_I) = \frac{40}{100} \times 10^5$ Pa = 0.4×10^5 Pa

Partial pressure of $I_2 (P_{I_2}) = \frac{60}{100} \times 10^5 Pa = 0.60 \times 10^5 Pa$

$$K_P = \frac{P_I^2}{P_{I_2}} = \frac{(0.4 \times 10^5)^2}{0.6 \times 10^5} = 2.67 \times 10^4 Pa$$

5. B

6. A, C, D

7. A, B, D

8. A, C, D

9. A, B, C

10. A, B, D

11. (A→q, s); (B→q, s); (C→q, s); (D→s, t)

12. (A→r, t); (B→s, t); (C→p, q, t); (D→p, t)

13. 8

14. 1
$k = k_1 \times k_2 = 10^{-2} \times 10^2 = 10^o = 1$

15. $N_2O_3(g) \rightleftharpoons 2NO_2(g)$

$$\alpha = \frac{D-d}{d} = \frac{D}{d} - 1$$

At A, α = 0. Hence $\frac{D}{d} = 1$

16. Suppose at equilibrium, $P_{CO_2} = P$

Then $P_{NH_3} = 2p$ atm

We are given that P + 2p = 3 atm

∴ P = 1 atm i.e., P_{CO_2} = 1 atm and P_{NH_3} = 2 atm

Hence $K_p = P_{NH_3}^2 \times P_{CO_2} = 2^2 \times 1 = 4$

17. 2

18. $pH = pK_a + \log \frac{[Salt]}{[Acid]}$

pK_b = 10 (given)

pK_a = 4

[Salt] = [Acid]

∴ $pH = pK_a = 4$

FIITJEE REVIEW TEST-03

(Batches: 921 Lot)

IIT – JEE, 2021

MATHEMATICS

Time: 1 Hour **Maximum Marks: 69**

INSTRUCTIONS

A. Question Paper Format

1. The question paper consists of four parts.
2. **Part-A – (01-05)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **only one is correct.**
3. **Part-A – (06-10)** contains **05 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **one or more is/are correct**.
4. **Part-B – (11-12)** contains **2 multiple choice questions**. Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct
5. **Part-C – (13-18)** contains **06 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.

B. Marking scheme:

(i). For each question in **Part A (2)**, you will be awarded **3 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened. In all other cases, minus one **(–1) mark** will be awarded.

(ii). For each question in **Part A (3),** you will be awarded **4 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks	:	+ 1	For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
Zero Marks	:	0	If none of the bubbles is darkened.
Negative Marks:		– 2	In all other cases.

For example, if [A], [C] and [D] are all the correct options for a question, darkening all these three will get +4 marks; darkening only [A] and [D] will get +2 marks; and darkening [A] and [B] will get –2 marks, as a wrong option is also darkened.

(iii). For each question in **Part B (4),** you will be awarded **8 marks** if you darken only the bubble corresponding to the correct answer and zero mark if no bubbles are darkened.

Partial Marks	:	+ 2	For darkening a bubble corresponding **to each correct option,** provided NO incorrect option is darkened.
Other Cases	:	0	In all other cases.

(iv). For each question in **Part C (5),** you will be awarded **+3 marks** for correct answer and **there will be no negative marking.**

Enrolment No. : | | | | | | | | | | | | | | | | | | | | |

Name : . . . . . .

Batch : . . . . . **Session :**. . . . . **Date:**. . . . . .

SECTION – III : MATHEMATICS

PART – A: (Single Correct Answer Type)

*This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.***

1. The value of $\lim_{n\to\infty} n^{-n^2}\left[(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)...\left(n+\frac{1}{2^{n-1}}\right)\right]^n$ is

   (A) e (B) e^2
   (C) e^3 (D) e^4

2. A derivable function $f : R^+ \to R$ satisfies the condition $f(x) - f(y) = \log_e x - \log_e y + x - y \ \forall\ x, y \in R^+$. If g(x) denotes the derivative of f(x). Then the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$ is

   (A) 5050 (B) 5510
   (C) 5150 (D) 1550

3. If f(x) is a function $f : R \to R$ and satisfies the property
   (i) if f(f(x)) = x for all real numbers x
   (ii) If f(– f(x)) = – x for all real numbers x
   Then number of linear functions have both property (i) and (ii)

   (A) 0 (B) 2
   (C) 3 (D) infinite

4. If the maximum distance of any point on the ellipse $x^2 + 2y^2 + 2xy = 1$ from its centre is r. Then r is equal to

   (A) $\frac{\sqrt{6}+1}{2}$ (B) $\frac{\sqrt{5}+1}{2}$
   (C) $\frac{\sqrt{3}+1}{2}$ (D) $\frac{\sqrt{2}+1}{2}$

5. Let C be a curve which is the locus of the point of intersection of lines x = 2 + m and my = 4 – m, A circle $(x – 2)^2 + (y + 1)^2 = 25$ intersects the curve C at four points P, Q, R and S. If O is the centre of the curve C, then $(OP)^2 + (OQ)^2 + (OR)^2 + (OS)^2$ is

   (A) 25 (B) 50
   (C) 100 (D) 200

Space for rough work

PART – A: (Multi Correct Answer Type)

*This section contains **05 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.*

6. If A and B are variable points on the ellipse $x^2 + 4y^2 = 4$ and O is origin. If the pair of variable straight lines $x^2 + 4y^2 + axy = 0$ represents equation OA & OB. Then locus of the point of intersection of tangents drawn at points A and B
   (A) $x - 2y = 0$ (B) $2x - y = 0$
   (C) $x + 2y = 0$ (D) $2x + y = 0$

7. Which of the following equations in parametric form can represent a hyperbola where 't' is a parameter
   (A) $x = \frac{a}{2}\left(t + \frac{1}{t}\right), y = \frac{b}{2}\left(t - \frac{1}{t}\right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0, \frac{x}{a} + \frac{ty}{b} = 1$
   (C) $x = e^t + e^{-t}$ and $y = e^t - e^{-t}$ (D) $x^2 - 6 = 2\cos t, y^2 + 2 = 4\cos^2\left(\frac{t}{2}\right)$

8. The graph of $f : R \to R$ be defined by $y = f(x)$ is symmetric with respect to $x = a$ and $x = b$. Which of the following is true
   (A) $f(2a - x) = f(x)$ (B) $f(2a + x) = f(-x)$
   (C) $f(2b + x) = f(-x)$ (D) f is periodic

9. The value of $\lim_{n\to\infty} \sum_{r=1}^{n} \cot^{-1}\left(\frac{r^3 - r + \frac{1}{r}}{2}\right)$ is λ/μ then
   (A) $\lambda = \pi$ (B) $\mu = 4$
   (C) $\mu = 2$ (D) $\lambda + \mu = \pi + 3$

10. A function f(x) satisfies the relation $f(x + y) = f(x) + f(y) + xy(x + y)\ \forall\ x, y \in R$ if $f'(0) = -1$, then
   (A) f(x) is polynomial (B) f(x) is an exponential
   (C) f(x) is twice differentiable for all $x \in R$ (D) $f'(3) = 8$

Space for rough work

PART – B: (Matrix Match)

*This section contains **2 multiple choice questions**. Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct*

11. Match the following

	Column – I		Column – II
(A)	The number of value of x in $(0, 2\pi)$ where the function $f(x) = \frac{\tan x + \cot x}{2} - \left\lvert\frac{\tan x - \cot x}{2}\right\rvert$ is continues but not differentiable is	(p)	2
(B)	The number of points where the function $f(x) = \min\{1, 1 + x^3, x^2 - 3x + 3\}$ is non-differentiable is	(q)	0
(C)	The number of points where $f(x) = (x + 4)^{1/3}$ is non-differentiable is	(r)	4
(D)	Consider $f(x) = \begin{cases} -\frac{\pi}{2}\log\left(\frac{2x}{\pi}\right) + \frac{\pi}{2}, & 0 < x \le \frac{\pi}{2} \\ \sin^{-1}(\sin x), & \frac{\pi}{2} < x \le \frac{3\pi}{2} \end{cases}$ number of points in $\left(0, \frac{3\pi}{2}\right)$ where f(x) is non-differentiable, is	(s)	1
		(t)	3

12. Match the following

	Column – I		Column – II
(A)	An ellipse is sliding along the co-ordinates axes. If the foci of the ellipse are (1, 1) and (3, 3). If area of the director circle of the ellipse is $\pi\lambda$, then λ is divisible by	(p)	3
(B)	If from a point $P(0, \lambda)$ two normals other than axes are drawn to the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, such that $\lvert\lambda\rvert < p$, then 4p is divisible by	(q)	4
(C)	For the hyperbola, $x^2 - 3y^2 = 9$, acute angle between its asymptotes is $\frac{\pi\lambda}{24}$, then λ is divisible by	(r)	5
(D)	If the midpoints of a chord of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ is (0, 3) and length of the chord is $\frac{4\lambda}{5}$, then λ is divisible by	(s)	6
		(t)	7

Space for rough work

PART – C: (Integer Answer Type)

*This section contains **06 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the OMR is to be bubbled.*

13. If $f(x)=\lim_{n\to\infty}\left(\sin^4 x+\frac{1}{4}\sin^4 2x+\frac{1}{4^2}\sin^4 2^2x+\ldots+\frac{1}{4^n}\sin^4\left(2^n x\right)\right)$ and g(x) is a differentiable function satisfying g(x) + f(x) = 1, then the maximum value of $\left(\sqrt{f(x)}+\sqrt{g(x)}\right)^4$ is _____

14. If $[\sin x]+\left[\frac{x}{2\pi}\right]+\left[\frac{2x}{5\pi}\right]=\frac{9x}{10\pi}$ where [.] denotes the greatest integer function, then the number of solutions in the interval (30, 60) is _____

15. Suppose the function f(x) – f(2x) has the derivative 5 at x = 1 and derivative 7 at x = 2. The derivative of the function f(x) – f(4x) – 10x at x = 1 is equal to _____

16. A tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ at any point P meet the line x = 0 at a point Q. Let R be the image of Q in the line y = x, then circle whose extremities of a diameter are Q and R passes through a fixed point (α, β). Then $\alpha + \beta$ is equal to _____

17. Let the major axis of a standard ellipse equals the trans verse axis of a standard hyperbola and their director circles have radius equal to 2R and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola respectively, then the value of $4e_2^2-e_1^2$

18. For each positive integer n, consider the point P in first quadrant with a abscissa n on the curve $y^2 – x^2 = 1$. If d_n represents the shortest distance from the point P to the line y = x and if $\lim_{n\to\infty}(n\cdot dn)=\lambda$, then $8\lambda^2$ is equal to

Space for rough work

FIITJEE REVIEW TEST

PHYSICS, CHEMISTRY & MATHEMATICS

ANSWERS

MATHEMATICS (SECTION –III)

PART – A

1. **B**	2. **C**	3. **B**	4. **B**
5. **C**	6. **A, C**	7. **A, C, D**	8. **A, B, C, D**
9. **A, C**	10. **A, C, D**		

11. **(A) → (r); (B) → (p); (C) → (s); (D) → (q)**
12. **(A) → (q); (B) → (p); (C) → (q); (D) → (q)**

PART – C

13. **4**	14. **2**	15. **9**	16. **0**
17. **6**	18. **1**		

HINTS & SOLUTIONS

MATHEMATICS (SECTION –III)

PART – A

1. **B**

$$= \lim_{n\to\infty}\left[\frac{(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right)\ldots\left(n+\frac{1}{2^{n-1}}\right)}{n^n}\right]^n$$

$$= e\cdot e^{1/2}\cdot e^{1/2^2}\cdot e^{1/(2)3}\ldots = e^{1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}\ldots\infty}$$

$$= e^{\frac{1}{1-\frac{1}{2}}} = e^2$$

2. **C**

$$f'(x) = \lim_{n\to 0}\frac{f(x+n)-f(x)}{n} = \frac{1}{x}+1$$

$g(x) = \frac{1}{x}+1$

$$\sum_{n=1}^{100} g\left(\frac{1}{n}\right) = g(1)+g\left(\frac{1}{2}\right)+g\left(\frac{1}{3}\right)+\ldots+g\left(\frac{1}{100}\right)$$

=(1 + 2 + 3 … + 100) + 100 = 5050 + 100

= 5150

3. **B**

Let f(x) = mx + b

(i) f(f(x)) = x

m(mx + b) + b = x

$m^2x + (m + 1)b = x$, $m = \pm 1$, $b = 0$

(ii) f(–f(x)) = – x

– f(x) . m + b = –x

$–m^2x + b(1 – n) = – x$

$m = \pm 1$ $b = 0$

so has two linear functions.

4. **B**

Centre is origin O(0, 0)

Let P be any point (r cosθ, r sinθ) on ellipse OP = r

$r^2\cos^2\theta + 2r^2\sin^2\theta + 2r^2\sin\theta\cos\theta = 1$

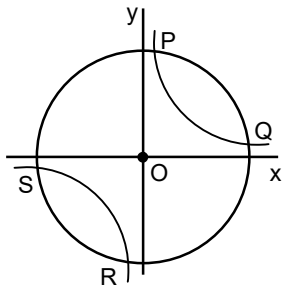
$$r^2 = \frac{2}{3+2\sin 2\theta-\cos 2\theta}$$

$$r^2_{max} = \frac{2}{3-\sqrt{5}} = \frac{2}{3-\sqrt{5}}\times\frac{3+\sqrt{5}}{3+\sqrt{5}}$$

$$= \frac{6+2\sqrt{5}}{4} = \left(\frac{\sqrt{5}+1}{2}\right)^2$$

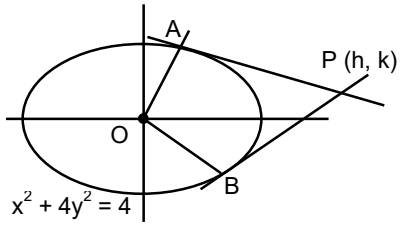
$$r_{max} = \frac{\sqrt{5}+1}{2}$$

5. **C**



Equation of curve is $(x - 2)(y + 1) = 4$
Let $x - 2 = X$, $y + 1 = Y$
Curve C is $XY = 4$
Equation of circle is $X^2 + Y^2 = 25 = r^2$
So, we know that
$OP^2 + OQ^2 + OR^2 + OS^2 = 4r^2$
$= 4 \times 25 = 100$

6. **A, C**



Equation of OA & OB $x^2 + 4y^2 + axy = 0$ … (i)

Equation of chord AB $\frac{xy}{4} + yk = 1$ … (ii)

Then equation of OA & OB

$$\frac{x^2}{4} + y^2 = \left(\frac{xh}{4} + yk\right)^2 \quad \text{… (iii)}$$

$$\left(\frac{1}{4} - \frac{h^2}{16}\right)x^2 + y^2\left(1 - k^2\right) - \frac{2xyhk}{4} = 0 \quad \text{… (iv)}$$

Equation (i) and (iv) are same so

$$\frac{4 - h^2}{16} = \frac{1 - k^2}{4}$$

$h^2 = 4k^2$, $x = \pm 2y$
$x \pm 2y = 0$

7. **A, C, D**

(A) hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(B) ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(C) hyperbola, $x^2 - y^2 = 4$

(D) hyperbola $x^2 - y^2 = 6$

8. **A, B, C, D**

Given $f(a + x) = f(a - x)$

(A) $f(2a - x) = f(a + a - x) = f(a - (a - x)) = f(x)$

(B) $f(2a + x) = f(-x)$

(C) $f(2b + x) = f(-x)$

(D) $f(2a + x) = f(2b + x)$ so periodic with period $2b - 2a$.

9. **A, C**

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \cot^{-1}\left(\frac{r^3 - r + \frac{1}{r}}{2}\right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \tan^{-1}\left(\frac{2r}{1 + r^4 - r^2}\right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \tan^{-1}\left(\frac{\left(r^2 + r\right) - \left(r^2 - r\right)}{1 + \left(r^2 + r\right)\left(r^2 - r\right)}\right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n} \tan^{-1}\left(r^2 + r\right) - \tan^{-1}\left(r^2 - r\right) = \frac{\pi}{2}$$

10. **A, C, D**

$$f'(x)=\lim_{n\to 0}\frac{f(x+n)-f(x)}{n},\ f(0)=0$$

$$=\lim_{n\to 0}\frac{f(n)}{n}+\frac{xn(x+n)}{n}$$

$$=\lim_{n\to 0}\frac{f(n)}{n}+\lim_{n\to 0}x(x+n)$$

$$f'(x)=-1+x^2$$

$$f(x)=\frac{x^3}{3}-x+c$$

$$f'(x)=\frac{3x^2}{3}-1=x^2-1$$

$f'(3) = 9 – 1 = 8$

11. **(A) → (r); (B) → (p); (C) → (s); (D) → (q)**

12. **(A) → (q); (B) → (p); (C) → (q); (D) → (q)**

PART – C

13. **4**

$$\sin^4 x=\sin^2 x-\frac{1}{4}\sin^2 2x$$

$$\frac{1}{4}\sin^4 2x=\frac{1}{4}\sin^2 2x-\frac{1}{4^2}\sin^2 2^2x$$

$$\frac{1}{4^n}\sin^4\left(2^n x\right)=\frac{1}{4^n}\sin^2\left(2^n x\right)-\frac{1}{4^{n+1}}\sin^2\left(2^{n+1}x\right)$$

So $f(x) = \sin^2 x$, $g(x) = \cos^2 x$

Max $\left(\sqrt{f(x)}+\sqrt{g(x)}\right)^4=4$

14. **2**

$$[\sin x]+\left[\frac{x}{2\pi}\right]+\left[\frac{2x}{5\pi}\right]=\frac{x}{2\pi}+\frac{2x}{5\pi}$$

$$[\sin x]=\frac{x}{2\pi}-\left[\frac{x}{2\pi}\right]+\frac{2x}{5\pi}-\left[\frac{2x}{5\pi}\right]$$

$$[\sin x]=\left\{\frac{x}{2\pi}\right\}+\left\{\frac{2x}{5\pi}\right\} \qquad \ldots (i)$$

RHS is $\left\{\frac{x}{2\pi}\right\}+\left\{\frac{2x}{5\pi}\right\}\geq 0$

While $[\sin x] = 0, 1, -1$ so $[\sin x] \neq -1$

So $[\sin x] = 0$, $\left\{\frac{x}{2\pi}\right\}+\left\{\frac{2x}{5\pi}\right\}=0$ if $x = 10\pi, 20\pi$.

15. **9**

Let $h(x) = f(x) – f(2x)$

$h'(x) = f'(x) – 2f'(2x)$

$f'(1) – 2f'(2) = 5$ … (i)

$f'(2) – 2f'(4) = 7$ … (ii)

Let $g(x) = f(x) – f(4x) – 10x$

$g'(x) = f'(x) – 4f'(4x) – 10$

= 19 – 10

g′(1) = 9.

16. **0**

Equation of tangent of the ellipse at P(5 cosθ, 4 sinθ) is $\frac{x}{5}\cos\theta + \frac{y}{4}\sin\theta = 1$

It meets the line x = 0 at Q(0, 4 cosecθ) and image of Q in the line y = x is R (4 cosec θ, 0)
Equation of the circle is

$x^2 + y^2 - 4(x + y)\text{ cosec }\theta = 0$

each member of the family passes through the intersection of $x^2 + y^2 = 0$ and x + y = 0
so fixed point (0, 0)

$\alpha + \beta = 0$.

17. **6**

Equation ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a > b)

And its director circles $x^2 + y^2 = a^2 + b^2$

$b^2 = a^2(1 - e_1^2) = 4R^2 \Rightarrow 2 - e_1^2 = \frac{4R^2}{a^2}$ … (i)

Equation of hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b_1^2} = 1$, director circle $x^2 + y^2 = a^2 - b^2$

$b_1^2 = a^2\left(e_2^2 - 1\right)$

$a^2 - b_1^2 = R^2 \Rightarrow a^2 - a^2(e_2^2 - 1) = R^2$

$2 - e_2^2 = \frac{R^2}{a^2}$ … (ii)

$2 - e_1^2 = 4\left(2 - e_2^2\right)$

$\Rightarrow \quad 4e_2^2 - e_1^2 = 6$

18. **1**

$P\left(n, \sqrt{1+n^2}\right)$

$d_n = \frac{\sqrt{1+n^2} - n}{\sqrt{2}}$

$\lambda = \lim_{n\to\infty} n\left(\frac{\sqrt{1+n^2} - n}{\sqrt{2}}\right)$

$\lim_{n\to\infty} \frac{n\left(1+n^2-n^2\right)}{\sqrt{2}\left(\sqrt{1+n^2}+n\right)} = \frac{1}{2\sqrt{2}} = 8\lambda^2 = 1.$